

BFS OUTPUT

```
Problems Output Debug Console Terminal Ports
@20525pcs002-ui →/workspaces/AI ML-RECORD (main) $ python bfs.py
BFS Traversal:
FireStation BuildingA BuildingB Floor1 Floor2 Floor3 @20525pcs002-ui →/workspaces/AI ML-RECORD (main) $
```

DFS OUTPUT

```
Problems Output Debug Console Terminal Ports
@20525pcs002-ui →/workspaces/AI ML-RECORD (main) $ python dfs.py
Following is the Depth First Search Traversal:
['Router', 'PC1', 'Printer', 'PC2', 'Server']
@20525pcs002-ui →/workspaces/AI ML-RECORD (main) $
```

INFORMED SEARCH OUTPUT

```
@20525pcs002-ui →/workspaces/AI ML-RECORD (main) $ python informed.py
Random Rabbits:
[Rabbit with 21 carrots., Rabbit with 23 carrots., Rabbit with 17 carrots., Rabbit with 50 carrots., Rabbit with 36 carrots.]

Carrots of First Rabbit: 21

Best Rabbit: Rabbit with 50 carrots.
```

NAÏVE BAYES OUTPUT

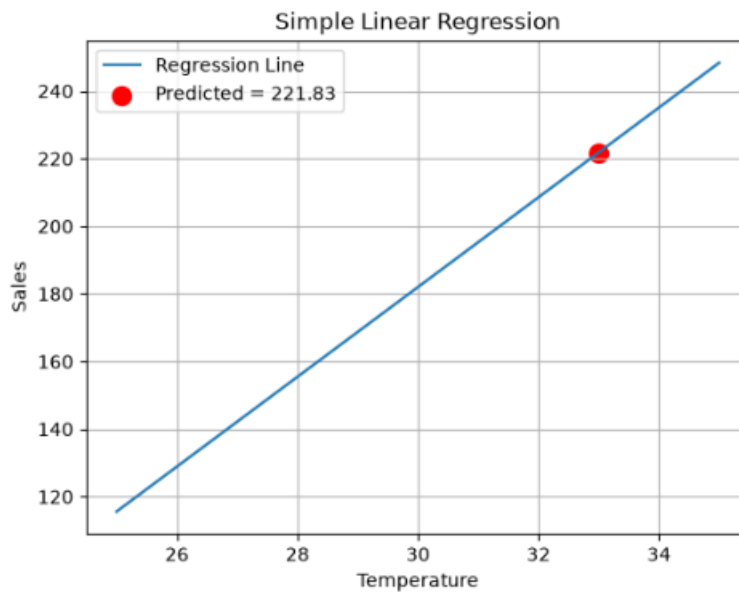
```
@20525pcs002-ui →/workspaces/AI ML-RECORD (main) $ python naive_bayes.py
Has good programming skills? (yes/no): yes
Selected = 0.82
Rejected = 0.18
Prediction: SELECTED
```

BAYESIAN NETWORKS OUTPUT

```
@20525pcs002-ui →/workspaces/AI ML-RECORD (main) $ python bayesian.py
CCTV Security
-----
P(Intruder) = 0.1
P(Motion|Intruder) = 0.95
P(Motion) = 0.2
Intruder Probability:
0.47
```

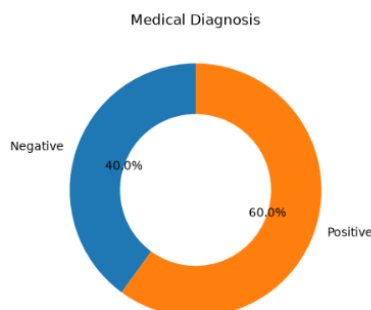
REGRESSION MODEL OUTPUT

```
@20525pcs002-ui →/workspaces/AIML-RECORD (main) $ python regression_model.py  
Predicted Sales: 221.83
```



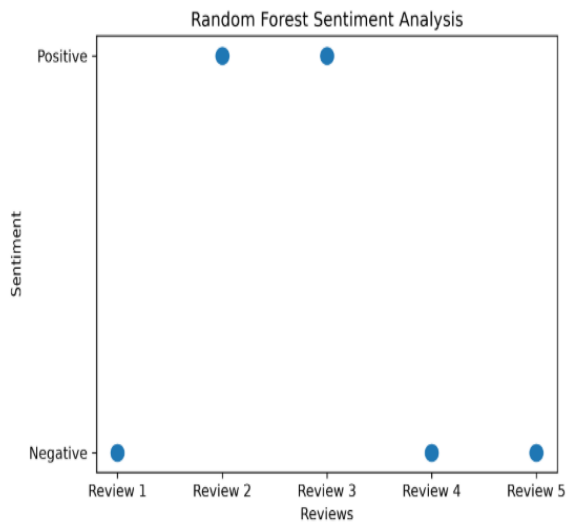
DECISION TREE OUTPUT

```
@20525pcs002-ui →/workspaces/AIML-RECORD (main) $ python decision_tree.py  
Diagnosis: Positive
```



RANDOM FOREST OUTPUT

```
@20525pcs002-ui →/workspaces/AIML-RECORD (main) $ python random_forest.py  
Predictions: ['Negative' 'Positive' 'Positive' 'Negative' 'Negative']
```



SVM MODEL OUTPUT

```
@20525pcs002-ui →/workspaces/AIML-RECORD (main) $ python svm_models.py
Enter weight: 50
Enter size: 65
Predicted fruit: Orange
```

ENSEMBLE LEARNING OUTPUT

```
@20525pcs002-ui →/workspaces/AIML-RECORD (main) $ python ensembling_learning.py
Training Data: [60, 70, 80, 90, 50]

Enter new student mark (q to stop): 80
Model 1: 80.0
Model 2: 70.0
Model 3: 75.0
Final Prediction: 75.0
Result: Pass

Enter new student mark (q to stop): q
Program stopped.
```

CLUSTERING OUTPUT

```
@20525pcs002-ui →/workspaces/AIML-RECORD (main) $ python cluster.py
```

```
Enter transaction amount (q to stop): 65  
Amount: 65.0 -> Low Transaction
```

```
Enter transaction amount (q to stop): 4  
Amount: 4.0 -> Low Transaction
```

```
Enter transaction amount (q to stop): 67  
Amount: 67.0 -> Low Transaction
```

```
Enter transaction amount (q to stop): 24  
Amount: 24.0 -> Low Transaction
```

```
Enter transaction amount (q to stop): q  
Monitoring stopped.
```

EM FOR BAYESIAN NETWORKS OUTPUT

```
@20525pcs002-ui →/workspaces/AIML-RECORD (main) $ python em_bayesian.py
```

```
Enter CGPA: 80
```

```
Enter Skill Score: 4
```

```
Enter Interview Score: 2
```

```
===== EM BAYESIAN NETWORK =====
```

```
## EM Analysis
```

```
Cluster 0 Probability : 1.0
```

```
Cluster 1 Probability : 0.0
```

```
Most Likely Cluster   : 0
```

```
## Bayesian Prediction
```

```
Selection Probability : 100.0 %
```

```
Rejection Probability : 0.0 %
```

```
## Decision
```

```
Candidate Status      : SELECTED
```

```
Reason : Good CGPA, Skill and Interview Score
```

```
Prediction Completed Successfully
```

NN MODELS OUTPUT

```
@20525pcs002-ui →/workspaces/AI ML-RECORD (main) $ python nn_models.py
/usr/local/python/3.14.2/lib/python3.14/site-packages/sklearn/neural_network/_multilayer_perceptron.py:785: ConvergenceWarning:
tochastic Optimizer: Maximum iterations (2000) reached and the optimization hasn't converged yet.
  warnings.warn(
Enter Refrigerator Temperature: 75
Enter Door Opening Count: 45
Enter Usage Hours: 54
Enter Outside Temperature: 63
Predicted Energy Consumption: LOW
```

DEEP LEARNING FOR NN MODELS

```
@20525pcs002-ui →/workspaces/AI ML-RECORD (main) $ python deep_nn.py
Enter Temperature: 65
Enter Humidity: 43

----- Cosmetic Product Risk Prediction -----
Opened Days : 60
Temperature : 65 °C
Humidity    : 43 %
Risk Level  : WARNING
```